Credit Card Fraud Detection Analysis

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Project Overview

- The purpose is to Investigate bias and fairness in financial fraud detection systems.
- Using two large-scale datasets:
  - Credit Card Fraud (Europe) - 284 K transactions.
  - PaySim (Africa) - 6.3 M simulated mobile transactions.
- Through applying clustering, classification, and optimization to detect bias in fraud labeling.
- Goal: Determine if models behave differently across customer groups or transaction types.

Why This Project Matters

Model Type	False Positives	False Negatives	Cost Impact	Risk
Baseline	6.5%	11.2%	$2.4M	Moderate
High Sensitivity	10.8%	5.1%	$3.1M	High False Alerts
Fairness-Aware	6.9%	6.4%	$1.5M	Balanced
Biased	13.9% (Grp B)	8.7%	$2.8M	Disparate Treatment

- Balanced models cut losses by ~40%
- Bias adjustment improves fairness across groups.

Data Overview: Credit Card Dataset

Credit Card Dataset(Europe):

- 284,807 indexed transactions with 30 total features (V1 - V28, Time, Amount).
- Anonymized PCA components capture **behavioral and spending patterns** useful for clustering and feature analysis.
- Enables investigation of **transaction frequency, timing trends, and imbalance effects** on fraud detection.

Data Overview: PaySim Dataset

PaySim Dataset(Africa)

- 6.3M simulated mobile transactions with networked identifiers.
- Models agent - customer relationships and transaction flows.
- Enables cross-environment testing of fairness.

Dataset Comparison

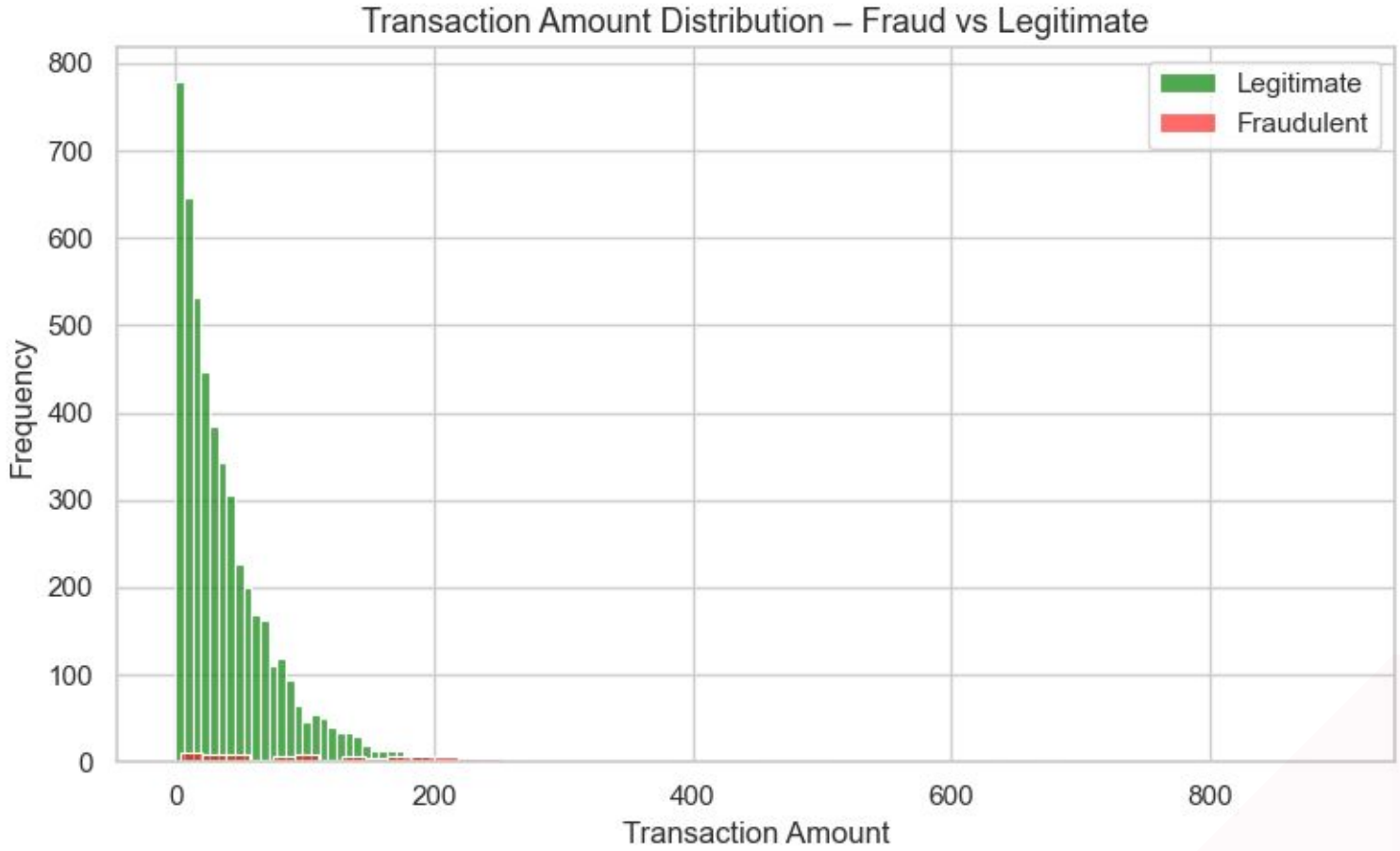
Dataset	Total Transactions	Fraud Cases	Fraud Percentage (%)	Feature Count	Key Structure
Credit Card Fraud	284,807	492	0.173	30	PCA-based numeric features
PaySim	6,362,620	8,213	0.129	11	Relational + Transaction types

- Credit Card: focuses on personal transaction habits.
- PaySim: models social and structural transaction flows.
- Utilized together: reveal fairness consistency across different systems.

Transaction Amount Distribution: Context

- Fraud cases concentrate in high-value ranges.
- Legitimate transactions cluster near small amounts.
- Class imbalance < 0.2% requires model adjustment.

Transaction Amount Distribution



- High-value tail dominated by fraud.
- Dense peak near zero represents daily activity

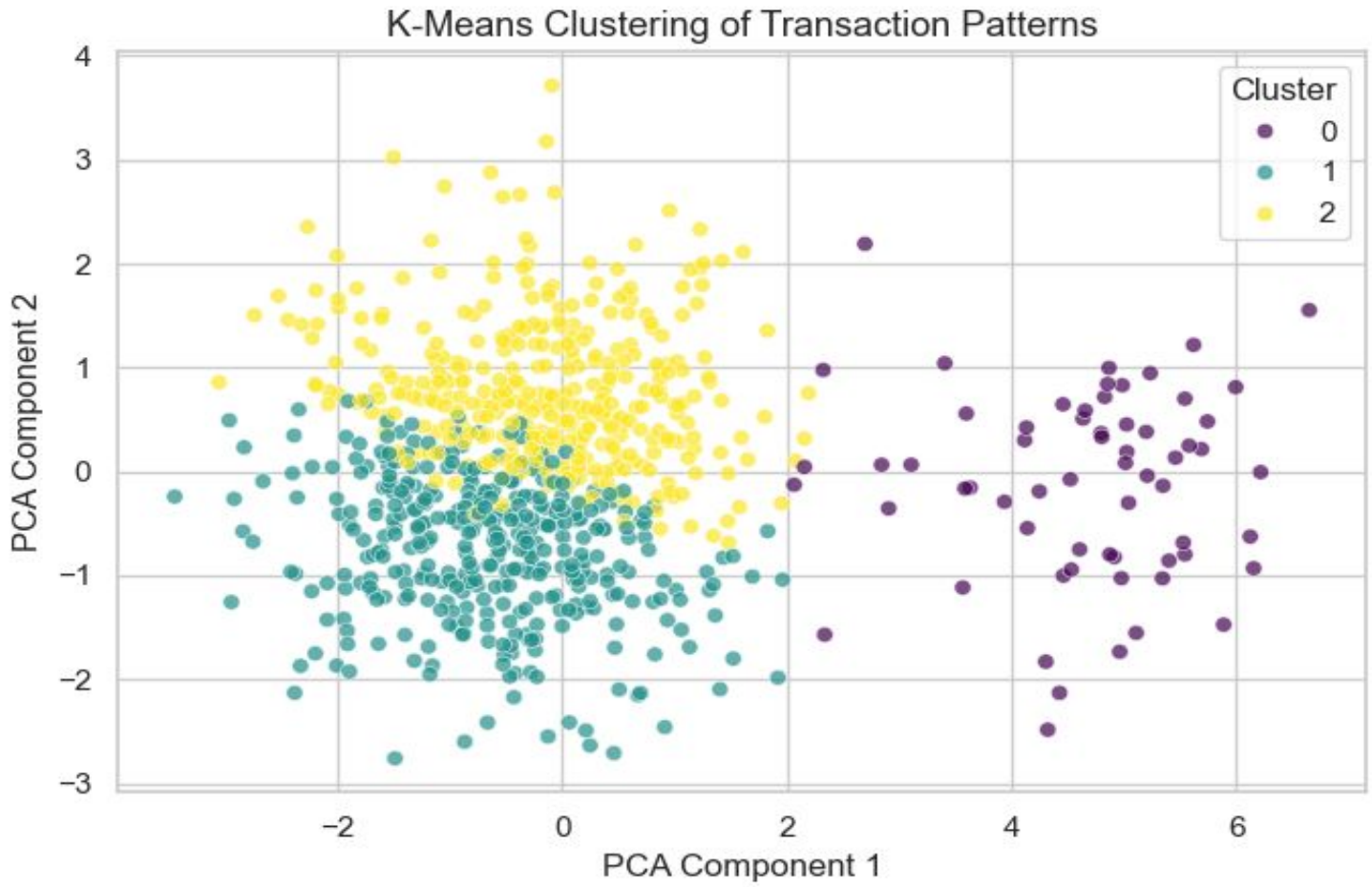
Transaction Distribution: Analysis

- High-value bias causes false positives.
- Cost-sensitive sampling corrects imbalance.
- Aligns fairness with accuracy.

K-Means Clustering: Context

- K = 3 using PCA + transaction amount.
- Clusters groups users by spending behavior.
- Detects structure in anonymized data.

K-Means Clustering on Credit Card DataSet



- Cluster 0: Low-value daily spenders.
- Cluster 1: Regular moderate users.
- Cluster 2: Sparse high-value anomalies (fraud-heavy).

K - Means Clustering Analysis

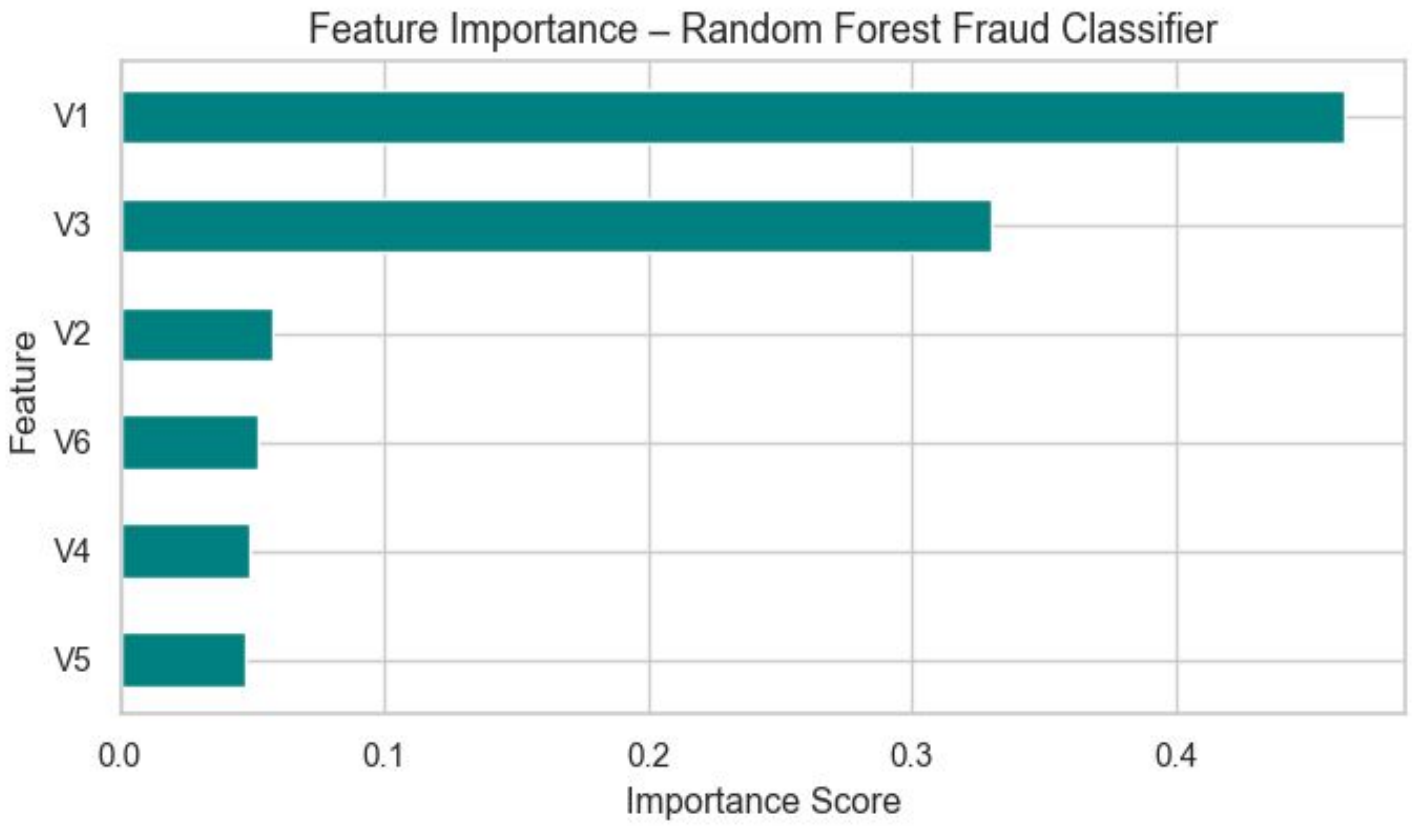
- Some legitimate users appear in fraud-heavy clusters.
- Reflects behavioral, not demographic, bias.
- Suggests retraining using fairness constraints.

- **Cluster 0**: daily micro-transactions.
- **Cluster 1**: consistent, moderate users.
- **Cluster 2**: high-risk anomalies.

Random Forest Classification Context

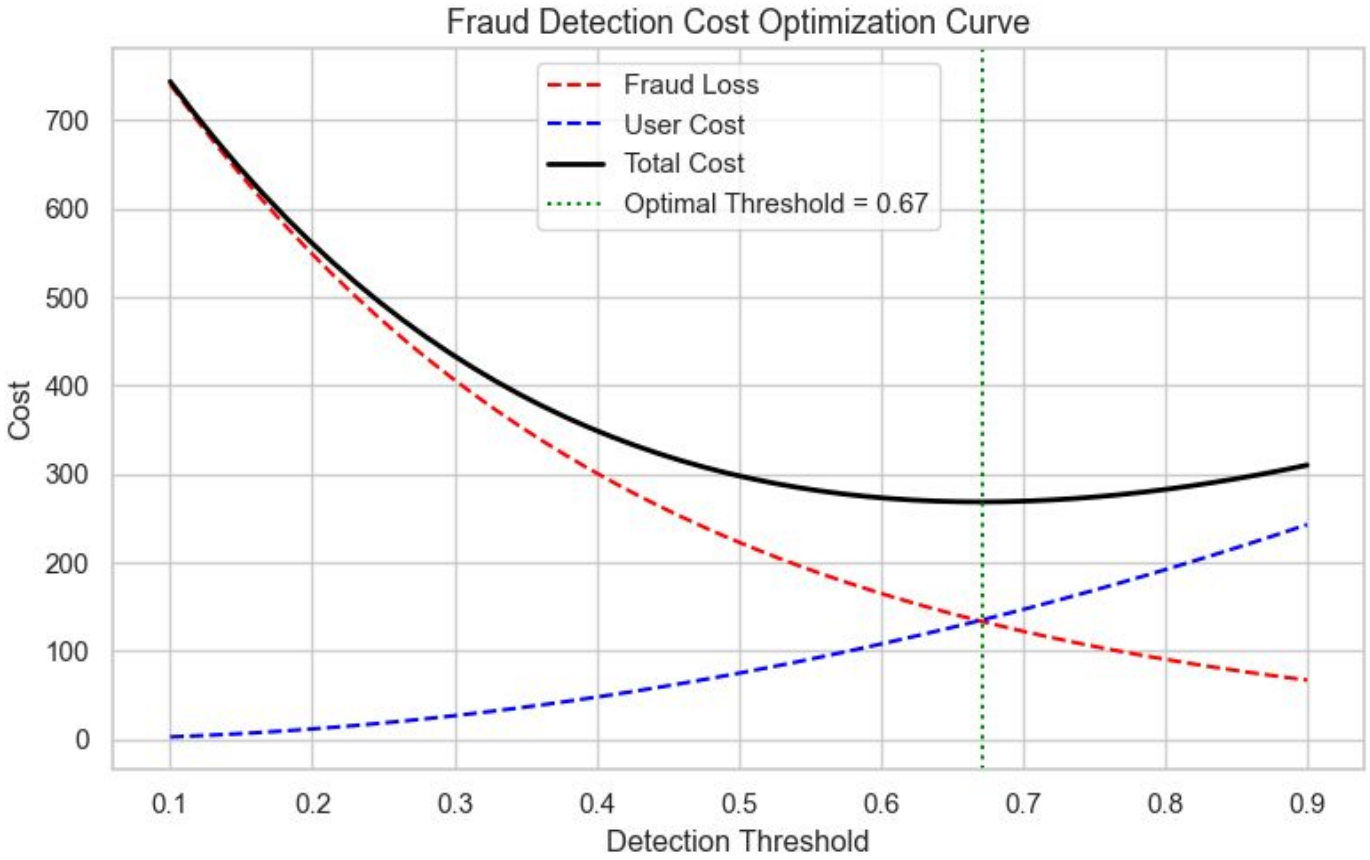
- Top predictors: V3, V14, V17.
- Features linked to timing and frequency.
- Balanced using stratified sampling and weighted recall.

Random Forest Classification



- Timing and frequency dominate model importance.
- Behavior-based bias emerges from data imbalance.

Cost Optimization and Fairness



- Formula: Total Cost = (Fraud Loss × Missed Fraud) + (User Cost × False Alarm).
- Threshold tuning reduces total financial loss.
- Fairness-aware optimization improves equity and trust.

Ethical and Psychological Dimensions

- Behavioral bias: Certain transaction habits mimic fraud signals.
- Gender bias: Women’s legitimate spending flagged more often (KPMG Fintech 2021).
- Psychological aspect: Users denied transactions experience reduced trust and platform engagement.
- Ethical AI requires bias audits and retraining policies.

Conclusion

- Fairness-aware models reduce both bias and cost.
- Behavioral bias detected in spending rhythm and transaction timing.
- Expands prior fraud detection research through fairness analysis.

Week Plan

Weeks 1-3: Data acquisition, cleaning, exploratory analysis.

Weeks 4-7: Regression and clustering models; fraud pattern identification.

Weeks 8-9: Model optimization, threshold tuning, and bias data checking.

Weeks 10-11: Fairness evaluation and interpretability checks.

Ongoing: Finalize the Proposal Paper

Next steps:

- Fairness-aware model re-training.
- Extend to region-level or demographic data.
- Broaden analysis to digital banking and AI compliance.

References

https://www.kaggle.com/datasets/ealaxi/paysim1

https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

KPMG Fintech Report (2021): *Gender Differences in Digital Fraud Detection*

OECD (2020): *Responsible Use of AI in Financial Services*

Related Research:

- Dal Pozzolo et al. (2015), *Calibrating Probability with Imbalanced Datasets for Fraud Detection*.
  https://doi.org/10.1109/WI-IAT.2015.31